

1) Inductor Selection and Calculation

1.1 Duty Cycle

Worst-case condition occurs at **maximum input voltage (20 V)**:

$$D = \frac{V_{out}}{V_{in}} = \frac{5}{20} = 0.25$$

1.2 Inductor Ripple Current

Inductor ripple current is typically chosen as **20–40% of output current**.

Chosen ripple:

$$\Delta I_L = 0.25 \times I_{out} = 0.25 \times 3 = 0.75 \text{ A}$$

1.3 Inductor Value Calculation

$$L = \frac{V_{out}(1 - D)}{\Delta I_L \cdot f_s}$$
$$L = \frac{5(1 - 0.25)}{0.75 \times 150000}$$
$$L = 33 \mu\text{H}$$

1.4 Inductor Current Rating

Peak inductor current:

$$I_{L,peak} = I_{out} + \frac{\Delta I_L}{2}$$
$$I_{L,peak} = 3 + 0.375 = 3.375 \text{ A}$$

For safe operation, inductor current rating $\geq 1.3\text{--}1.5 \times I_{out}$.

$$\text{Current rating} \geq 4\text{A}$$

2) Diode Selection

A buck converter switches at high frequency, a Schottky diode is used since it has very fast switching (no reverse-recovery delay) and a low forward voltage drop, which reduces power loss, heat, and improves efficiency compared to a normal diode.

2.1 Diode current rating

Duty cycle at worst-case input voltage:

$$D = \frac{V_{out}}{V_{in(max)}} = \frac{5}{20} = 0.25$$

Average diode current:

$$I_{D(avg)} = I_{out} \times (1 - D)$$
$$I_{D(avg)} = 3 \times (1 - 0.25) = 2.25A$$

Current rating $\geq 5A$

2.2 Diode Voltage Rating

Worst-case reverse voltage across diode:

$$V_{in(max)} = 20V$$

$$V_{D,rated} \geq 2 \times V_{in(max)} = 40V$$

3) Capacitor (Cout) Selection

Output ripple voltage limit:

$$\Delta V_{ESR(max)} = 50 \text{ mV} = 0.05 \text{ V}$$

3.1 ESR Calculation:

$$ESR_{max} = \frac{\Delta V_{ESR}}{\Delta I_L}$$
$$ESR_{max} = \frac{0.05}{0.75}$$
$$ESR_{max} = 0.0667 \Omega \approx 67 \text{ m}\Omega$$

To account for both ESR and capacitive ripple, the ripple budget is split approximately equally.

$$\Delta V_C = \frac{50}{2} = 25 \text{ mV} = 0.025 \text{ V}$$

3.2 The capacitive ripple voltage for a buck converter is given by:

$$C \geq \frac{\Delta I_L}{8 \cdot f_s \cdot \Delta V_C}$$
$$C \geq \frac{0.75}{30000}$$
$$C \geq 250 \mu F$$

4) Capacitor (Cin) Selection

4.1 Capacitance calculation

The input ripple voltage is given by:

$$\Delta V_{in} = \frac{I_{out} \cdot D}{f_s \cdot C_{in}}$$

Rearranging to find the minimum input capacitance:

$$C_{in} \geq \frac{I_{out} \cdot D}{f_s \cdot \Delta V_{in}}$$

$$C_{in} \geq \frac{0.75}{75000}$$

$$C_{in} \geq 10 \mu F$$

4.2 Capacitor Voltage Rating

Maximum input voltage:

$$V_{in(max)} = 20V$$

Applying a 1.5× derating factor:

$$V_{rated} \geq 1.5 \times V_{in(max)}$$
$$V_{rated} \geq 1.5 \times 20 = 30V$$